

ETS052 Computer Communication (page references from Forouzan)

Chapter 1: Introduction (pages 3-21)

Chapter 2: Network models (pages 27-50)

Chapter 3: Data and Signals (pages 57-85, 89-94)

3.5: Only the introduction

Chapter 4.1-2: Digital Transmission (pages 106-110, 120-129)

4.1: Only introduction, *NRZ-L*, *Manchester*, and *Differential Manchester*

4.2: Exclude *Delta Modulation*

Chapter 5.1: Digital-to-Analog conversion (pages 141-152)

Exclude *Multilevel ASK*, *MFSK*, and *QPSK*

Chapter 6: Bandwidth Utilization: Multiplexing and Spreading (pages 161-185)

6.1 FDM: Exclude *The Analog Carrier System*

6.1 Synchronous TDM: Only *Time slots and frames*, and *Frame synchronizing*

Chapter 7: Transmission Media (pages 191-208)

Chapter 8.1-8.3: Switching (pages 213-227)

Chapter 10: Error Detection and Correction (pages 267-274, 277-280, 284-301)

10.2: Exclude *Hamming Distance* and *Minimum Hamming Distance*

10.3: Only parity-check code

Chapter 11: Data Link Control (pages 307-357)

11.4: Read only as an introduction to 11.5

11.6: See lecture notes

11.7: See lecture notes

Chapter 12: Multiple Access (pages 363-386)

12.6 CDMA: Only introduction, *Analogy* and *Idea*

Chapter 13: Wired LANs: Ethernet (pages 395-403, 406-416)

13.2: Exclude *10Base5*, *10Base2*, *10Base-T*, *10Base-F*

13.4: Exclude *Encoding*

13.5: Only introductions on Gigabit Ethernet and Ten-Gigabit Ethernet

Chapter 14: Wireless LANs (pages 421-437)

14.1: Only introduction, *Basic Service Set*, *DCF*, *Hidden and Exposed Station problems*, and *Physical layer*

14.2: Only *Architecture*, *Bluetooth Layers*, *Radio Layer*, and introduction to *Baseband Layer*

Chapter 15: Connecting LANs, Backbone Networks, and Virtual LANs (pages 445-463)

15.1: Exclude *Spanning Tree*, *Source Routing Bridges*, *Two-layer Switches*, and *Three-layer Switches*

Chapter 19: Network Layer: Logical Addressing (pages 549-558, 563-568)

19.1: Exclude *Subnetting*, *Supernetting*, *Address Depletion*, and all sections about *Hierarchy*

19.2: Exclude *Unicast*, *Multicast*, *Anycast*, *Reserved* and *Local Addresses*

Chapter 20: Internet Protocol (pages 579-594, 596-597)

20.2: Exclude *Options*

20.3: Only the introduction and *Advantages*

Chapter 21.1-2: Address Mapping, Error Reporting, and Multicasting (pages 611-617, 620-630)

21.1: Exclude *Proxy ARP*, *RARP*, and *BOOTP*

Chapter 22: Delivery, Forwarding, and Routing (pages 647-652, 655-664, 666-671, 674-676, 678-693)

22.2: Exclude *Address Aggregation*, *Longest Mask Matching*, *Hierarchical routing*, and *Geographical routing*

22.3: Exclude *Two-node instability*, *Three-node instability*, *RIP*, *OSPF*, *BGP*

22.4: Only *Unicast*, *Multicast* and *Broadcast*, *Multicast routing*, *Reverse Path Forwarding*

Chapter 23.1-3: UDP and TCP (pages 703-728)

23.3: Exclude *Flow control* and *Error control*

Chapter 24.2 Congestion (pages 763-765)

Chapter 25.1-4: Domain Name System (pages 797-806)

25.2: Exclude *FQDN*, *PQDN*

25.3: Exclude *Zone*, *Root server*, *Primary* and *Secondary Servers*

25.4: Exclude *Inverse Domains*

Chapter 31: Network Security (pages 961-990)

Content defined by lecture notes

Chapter 32: Security in the Internet: IPSec, SSL/TLS, PGP, VPN, and Firewalls (pages 995-1002, 1008, 1014-1015, 1021-1023)

32.1: Only *Two modes* and *Two Security Protocols*

32.2: Only the introduction

32.3: Only the introduction

SUNET and LUNET: See Lecture notes

Problem solving classes

Class 1: Error detection

Class 2: Flow and Error control + Multiplexing

Class 3: Ethernet

Class 4: IP

Class 5: TCP and UDP

Class 6: Routing

Laboratory Sessions

Lab 1: PPP

Lab 2: Networking